

MADISON EMSHOUSEN

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PROFESSIONAL SUMMARY

Insider Threat Investigator and ML Engineer with 3+ years of experience conducting end-to-end security investigations, building machine learning pipelines, and automating forensic workflows at enterprise scale. Proven ability to correlate endpoint, identity, and data-loss signals across CrowdStrike, Microsoft Purview, and Defender to reconstruct attacker and insider activity. Adept at translating raw data into clear, evidenced findings for legal, HR, and security leadership.

TECHNICAL SKILLS

Languages: Python, SQL, KQL, PowerShell, C++, Java, JavaScript, C#, R, Lua, HTML
ML / AI: Machine Learning, Large Language Models (LLMs), Predictive Modeling, Anomaly Detection, NLP, MLOps
Data & Analytics: ETL Pipelines, Data Engineering, Dashboard Development, Data Visualization, Matplotlib, Pandas
Databases: SQL (advanced), RDBMS, Query Optimization, Indexing, React.js
Security & Investigations: Insider Threat Investigations, Data Loss Prevention, Digital Forensics, Fraud Detection, Incident Response, OSINT, Chain of Custody, Threat Analysis
Security Tools: CrowdStrike Falcon, Microsoft Purview, Microsoft Defender, Microsoft Entra ID, Intune, Power Automate
Tools & Platforms: Jupyter, Git/GitHub, Tableau/BI tools, Agile

EXPERIENCE

Allstate Aug 2026 – Present
Insider Threat Investigator · Remote

- Investigate potential insider threats — including data exfiltration, policy violations, and access misuse — from first alert through documented findings, correlating endpoint telemetry from CrowdStrike Falcon and Microsoft Defender with Microsoft Purview data-loss and insider-risk signals.
- Write KQL queries to hunt across sign-in, device, and data-movement logs; automate evidence collection and case triage with PowerShell and Power Automate, eliminating repetitive manual steps.
- Use Microsoft Entra ID and Intune to confirm identity, access, and device posture; write up findings with a clear chain of evidence for HR, legal, and security leadership.

Control Risks @ Meta Oct 2024 – Aug 2026
Data Analyst — Investigation Enablement, Global Security & Investigations Remote

- Engineered end-to-end ML pipelines in Python and SQL to track and recover over **\$200M** in lost and stolen Meta devices, enabling real-time asset recovery and reducing financial exposure at enterprise scale.
- Conducted forensic SQL investigations into organized retail theft incidents, identifying repeat offenders across multiple break-in events by cross-referencing behavioral patterns and transaction records in complex Meta databases.
- Designed and deployed LLM-powered dashboards to surface trends and generate actionable intelligence, reducing manual analyst review time and improving investigative throughput.
- Automated data collection and end-to-end investigative workflows using custom Python scripts, eliminating repetitive manual processes and accelerating case resolution time.
- Collaborated with cross-functional teams including legal, security operations, and engineering to implement scalable, data-driven solutions supporting global investigations.
- Presented findings and intelligence reports to senior stakeholders, driving informed decision-making on asset protection strategy.

PROJECTS

Online Music Database | *SQL, Java, React.js*

- Designed and implemented a scalable RDBMS with optimized backend architecture supporting music playback, playlist management, and user interaction; applied advanced indexing and query optimization techniques to maximize performance under load.

Project Helios | *Python, Matplotlib, Data Visualization*

- Led a data visualization and analysis project using Python and Matplotlib to process solar flare datasets, uncover temporal trends, and deliver interactive visual reports with actionable insights.

EDUCATION

University of Houston Class of 2024
Bachelor of Science in Computer Science — Houston, Texas

ACHIEVEMENTS

🏆 CodeRed UH Hackathon Winner — University of Houston 2025
CodeCoogs Team Lead — Led student engineering organization 2024
CodeCoogs Member — Active contributor to CS community 2022 – 2024

VOLUNTEER EXPERIENCE

Computer Science Association — University of Houston Aug 2025 – Present

- Director of Marketing & Mentor — led HackHCC, HCC's first-ever hackathon; organize workshops and mentor CS students.